

## AMBERLEY, Australia

WMO: 945680

Lat: 27.6297S

Lon: 152.7111E

Elev: 83

StdP: 14.65

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 7             | 33.7       | 36.4 | 19.8                          | 14.9 | 70.8 | 24.8 | 18.9 | 64.1 | 21.2                  | 66.3 | 19.5 | 65.7 | 1.6                   | 190  | 0.376 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 1             | 19.7                   | 95.3            | 71.5 | 91.8 | 71.4 | 89.5 | 70.7 | 76.6                | 87.1 | 75.4 | 85.1 | 74.4 | 83.3 | 10.8                 | 60   |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 73.8                            | 126.3 | 79.5 | 73.1 | 123.1 | 78.9 | 71.7 | 117.6 | 78.1 | 40.1          | 87.7 | 39.0 | 85.2 | 37.9 | 83.0 | 82.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |       |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|-------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |       |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |       |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |       |
| 20.2              | 18.3 | 16.5 | DB                         | 28.8 | 103.5              | 2.1 | 3.6                                                | 27.3 | 106.1      | 26.1 | 108.2      | 24.9 | 110.3      | 23.4 | 112.9 |
|                   |      |      | WB                         | 28.3 | 79.5               | 2.4 | 1.4                                                | 26.6 | 80.5       | 25.2 | 81.3       | 23.9 | 82.0       | 22.2 | 83.1  |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 67.7        | 77.8  | 76.7  | 74.3  | 68.5  | 62.1  | 57.7  | 56.1  | 57.9  | 64.2  | 68.9  | 72.9  | 75.8  | (6)  |
|                                                                        | DBStd       | 8.74        | 3.97  | 3.84  | 3.57  | 3.67  | 4.27  | 4.52  | 4.39  | 4.53  | 4.73  | 4.66  | 4.65  | 4.48  | (7)  |
|                                                                        | HDD50       | 6           | 0     | 0     | 0     | 0     | 0     | 2     | 3     | 1     | 0     | 0     | 0     | 0     | (8)  |
|                                                                        | HDD65       | 930         | 0     | 0     | 0     | 10    | 109   | 221   | 279   | 224   | 71    | 14    | 2     | 0     | (9)  |
|                                                                        | CDD50       | 6466        | 862   | 748   | 753   | 555   | 377   | 233   | 192   | 248   | 425   | 585   | 687   | 801   | (10) |
|                                                                        | CDD65       | 1912        | 397   | 328   | 288   | 115   | 20    | 2     | 2     | 6     | 46    | 134   | 238   | 336   | (11) |
|                                                                        | CDH74       | 17147       | 3615  | 2797  | 2195  | 940   | 239   | 28    | 39    | 195   | 749   | 1367  | 2047  | 2936  | (12) |
|                                                                        | CDH80       | 5998        | 1454  | 1043  | 675   | 171   | 19    | 1     | 2     | 34    | 211   | 463   | 762   | 1163  | (13) |
| (14) Wind                                                              | WSAvg       | 6.4         | 7.6   | 7.4   | 6.7   | 5.5   | 4.8   | 5.2   | 5.3   | 5.8   | 6.5   | 7.1   | 7.4   | 7.5   | (14) |
| (15) Precipitation                                                     | PrecAvg     | 33.8        | 4.3   | 4.2   | 3.1   | 2.3   | 2.3   | 1.6   | 1.6   | 1.2   | 1.2   | 2.8   | 3.5   | 4.7   | (15) |
|                                                                        | PrecMax     | 55.0        | 25.0  | 11.7  | 10.1  | 14.3  | 16.9  | 15.3  | 8.4   | 3.6   | 4.1   | 12.6  | 10.6  | 15.5  | (16) |
|                                                                        | PrecMin     | 15.9        | 0.0   | 0.7   | 0.3   | 0.1   | 0.2   | 0.0   | 0.0   | 0.0   | 0.0   | 0.3   | 0.4   | 1.3   | (17) |
|                                                                        | PrecStd     | 8.1         | 3.9   | 2.7   | 2.2   | 3.0   | 3.0   | 2.5   | 1.7   | 0.9   | 1.1   | 2.1   | 2.2   | 2.6   | (18) |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 100.5 | 97.1  | 93.6  | 87.9  | 82.3  | 77.2  | 78.4  | 83.9  | 91.4  | 95.2  | 97.2  | 99.0  | (19) |
|                                                                        |             | MCWB        | 73.8  | 74.4  | 72.5  | 65.9  | 62.5  | 61.7  | 58.3  | 60.2  | 63.3  | 65.2  | 68.0  | 70.0  | (20) |
|                                                                        | 2%          | DB          | 94.2  | 91.9  | 89.3  | 84.3  | 78.9  | 73.8  | 74.2  | 78.9  | 85.6  | 89.3  | 91.8  | 93.3  | (21) |
|                                                                        |             | MCWB        | 73.2  | 73.7  | 72.1  | 66.6  | 61.7  | 59.7  | 57.5  | 58.4  | 62.4  | 64.9  | 69.2  | 72.0  | (22) |
|                                                                        | 5%          | DB          | 90.1  | 89.2  | 86.2  | 82.0  | 76.7  | 71.8  | 71.7  | 75.3  | 81.6  | 84.9  | 87.8  | 89.5  | (23) |
|                                                                        |             | MCWB        | 72.7  | 72.8  | 70.4  | 66.2  | 61.3  | 58.9  | 56.7  | 57.2  | 61.5  | 64.9  | 68.7  | 71.6  | (24) |
|                                                                        | 10%         | DB          | 87.5  | 86.1  | 83.8  | 79.2  | 74.3  | 69.8  | 69.5  | 72.1  | 78.3  | 81.1  | 84.3  | 86.2  | (25) |
|                                                                        |             | MCWB        | 72.1  | 71.8  | 69.5  | 65.2  | 61.0  | 58.3  | 55.8  | 56.1  | 60.4  | 64.3  | 67.6  | 70.5  | (26) |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 78.1  | 78.7  | 76.4  | 72.5  | 69.5  | 65.7  | 65.7  | 65.3  | 68.8  | 72.5  | 74.7  | 77.0  | (27) |
|                                                                        |             | MCDB        | 88.8  | 90.8  | 85.3  | 79.4  | 73.3  | 70.9  | 72.3  | 76.3  | 80.6  | 83.8  | 87.4  | 89.9  | (28) |
|                                                                        | 2%          | WB          | 76.5  | 76.6  | 74.6  | 70.2  | 66.9  | 63.8  | 62.5  | 62.9  | 66.7  | 69.9  | 72.8  | 75.3  | (29) |
|                                                                        |             | MCDB        | 86.8  | 86.8  | 82.6  | 77.8  | 73.5  | 69.4  | 67.7  | 71.7  | 77.4  | 80.3  | 84.1  | 86.5  | (30) |
|                                                                        | 5%          | WB          | 75.3  | 75.1  | 73.4  | 69.0  | 65.1  | 62.3  | 60.3  | 60.9  | 65.1  | 68.3  | 71.4  | 74.1  | (31) |
|                                                                        |             | MCDB        | 84.5  | 84.1  | 81.0  | 76.8  | 71.9  | 68.3  | 67.0  | 69.3  | 75.1  | 78.2  | 81.5  | 83.8  | (32) |
|                                                                        | 10%         | WB          | 74.2  | 73.8  | 72.2  | 67.6  | 63.6  | 60.7  | 58.4  | 59.2  | 63.6  | 66.9  | 70.0  | 72.7  | (33) |
|                                                                        |             | MCDB        | 82.9  | 81.7  | 79.5  | 75.6  | 70.5  | 66.8  | 65.7  | 68.0  | 73.2  | 76.4  | 79.5  | 81.3  | (34) |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 19.7  | 18.9  | 19.8  | 23.2  | 26.0  | 25.3  | 28.5  | 30.8  | 29.1  | 25.3  | 22.2  | 20.8  | (35) |
|                                                                        |             | MCDBR       | 24.5  | 23.7  | 24.1  | 26.4  | 29.3  | 28.1  | 31.8  | 35.2  | 34.8  | 31.8  | 28.2  | 26.3  | (36) |
|                                                                        | 5% WB       | MCWBR       | 7.9   | 7.6   | 8.3   | 10.9  | 13.9  | 15.0  | 16.5  | 16.8  | 14.2  | 12.7  | 9.8   | 9.1   | (37) |
|                                                                        |             | MCWBR       | 20.3  | 20.0  | 18.9  | 21.3  | 20.7  | 20.7  | 23.2  | 26.9  | 27.3  | 26.0  | 23.3  | 22.0  | (38) |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.390 | 0.378 | 0.357 | 0.338 | 0.314 | 0.309 | 0.299 | 0.313 | 0.355 | 0.378 | 0.384 | 0.397 | (40) |
|                                                                        |             | taud        | 2.482 | 2.524 | 2.580 | 2.607 | 2.627 | 2.633 | 2.649 | 2.585 | 2.464 | 2.435 | 2.457 | 2.446 | (41) |
|                                                                        | Ebn at Noon |             | 302   | 301   | 298   | 289   | 282   | 277   | 285   | 293   | 293   | 297   | 302   | 301   | (42) |
|                                                                        |             | Edh at Noon | 37    | 35    | 31    | 28    | 24    | 23    | 23    | 28    | 34    | 37    | 38    | 39    | (43) |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 2055  | 1880  | 1626  | 1415  | 1181  | 978   | 1132  | 1416  | 1735  | 1923  | 2085  | 2070  | (44) |
|                                                                        | RadStd      |             | 175   | 169   | 96    | 71    | 66    | 70    | 89    | 108   | 141   | 185   | 193   | 219   | (45) |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |      |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |      |
|      |                         | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |      |
| (46) | Station Only            | +0.79 | N/A     | N/A    | N/A     | N/A   | N/A   | -4          | -121  | +367  | +243  | (46) |
| (47) | Regional (28 neighbors) | +0.74 | +0.94   | N/A    | N/A     | N/A   | N/A   | N/A         | -121  | +335  | +202  | (47) |

Nomenclature: See separate page